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**Subject: Cell Mol**

**Section: A**

**Cell & Molecular Biology**

**Lab Activity:**

**Characterization of a Plastid Genome**

Cell and Molecular Biology

**Data Source and Genome Selection**

Organism: *Musa itinerans*

Family: Musaceae

NCBI Accession/Version: NC\_035723.1

Database Source: NCBI Reference Sequence (RefSeq)

Genome length: 168,985 bp

Topology: Circular

Associated reference/publication: *The complete chloroplast genome sequence of the Musa itinerans* — Li, W., Liu, Y., and Gao, L.Z. Files to Obtain

**Required Plastid Genome Characterization**

| Characteristic             | Result                                                                                                                                        |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Chosen genus and species   | <i>Musa itinerans</i>                                                                                                                         |
| Family                     | Musaceae                                                                                                                                      |
| NCBI accession/version     | NC_035723.1                                                                                                                                   |
| Complete genome size       | 168,985 bp                                                                                                                                    |
| GC content                 | 36.82%                                                                                                                                        |
| Genome topology            | Circular                                                                                                                                      |
| LSC size                   | 87,828 bp                                                                                                                                     |
| SSC size                   | 11,329 bp                                                                                                                                     |
| IR size                    | 34,914 bp each                                                                                                                                |
| Total annotated genes      | 133                                                                                                                                           |
| Protein-coding genes (CDS) | 88                                                                                                                                            |
| tRNA genes                 | 37                                                                                                                                            |
| rRNA genes                 | 8                                                                                                                                             |
| Introns                    | Present; examples include <i>rps16</i> , <i>atpF</i> , <i>rpoC1</i> , <i>ycf3</i> , <i>clpP</i> , <i>petB</i> , <i>petD</i> , and <i>ndhA</i> |
| Pseudogenes                | None annotated                                                                                                                                |
| Gene duplications          | Present, mainly genes located within the inverted-repeat (IR) regions                                                                         |
| Sequence records in Galaxy | 1                                                                                                                                             |

## Gene groups to identify

| Gene group               | Genes found                                                                              |
|--------------------------|------------------------------------------------------------------------------------------|
| Photosystem I (psa)      | psaA, psaB, psaC, psaI, psaJ                                                             |
| Photosystem II (psb)     | psbA, psbB, psbC, psbD, psbE, psbF, psbH, psbI, psbJ, psbK, psbL, psbM, psbN, psbT, psbZ |
| ATP synthase (atp)       | atpA, atpB, atpE, atpF, atpH, atpI                                                       |
| Cytochrome b6f (pet)     | petA, petB, petD, petG, petL, petN                                                       |
| Carbon fixation          | rbcL                                                                                     |
| RNA polymerase (rpo)     | rpoA, rpoB, rpoC1, rpoC2                                                                 |
| Ribosomal proteins (rpl) | rpl2, rpl14, rpl20, rpl22, rpl23, rpl32, rpl33, rpl36                                    |
| Ribosomal proteins (rps) | rps2, rps3, rps4, rps7, rps8, rps11, rps12, rps14, rps15, rps16, rps18, rps19            |
| rRNA (rrn)               | rrn4.5, rrn5, rrn16, rrn23                                                               |
| tRNA (trn)               | Multiple trn genes; 37 tRNA features are annotated                                       |
| Other conserved genes    | matK, clpP, accD, cemA, ycf1, ycf2, ycf3, ycf4                                           |

## Questions for the Student Report

1. Give the full scientific name, family, NCBI accession/version, database source, and complete plastid-genome size of your selected organism.
  - The selected organism is *Musa itinerans*, a member of the family Musaceae. Its complete chloroplast genome is available in the NCBI Reference Sequence (RefSeq) database under accession NC\_035723.1. The complete plastid genome is 168,985 bp long. The genome is circular.
2. What evidence shows that the sequence is a complete plastid/chloroplast genome rather than a barcode marker, genome fragment, or nuclear sequence?
  - The NCBI record is explicitly identified as *Musa itinerans chloroplast, complete genome* and has the RefSeq accession NC\_035723.1. The sequence is 168,985 bp, is reported as circular, and contains the typical chloroplast genome organization with LSC, SSC, and two IR regions. These features demonstrate that it is a complete chloroplast genome rather than a barcode gene, genome fragment, or nuclear sequence.
3. Describe the overall organization of the plastid genome. Does it contain the common LSC-IR-SSCIR arrangement? Give the sizes of these regions when available.
  - The LSC (Large Single-Copy) region is 87,828 bp, the SSC (Small Single-Copy) region is 11,329 bp, and each IR (Inverted Repeat) is 34,914 bp. The chloroplast genome follows the typical arrangement: LSC – IR – SSC – IR

4. Summarize the annotated gene content: total genes, protein-coding genes, tRNA genes, rRNA genes, and pseudogenes. Explain why genes located in the inverted-repeat regions may appear in two copies.
  - The current NCBI RefSeq annotation contains 133 gene features, including 88 protein-coding sequences (CDS), 37 tRNA genes, and 8 rRNA genes. No pseudogenes are annotated in the current record. The original publication reported 113 unique genes, consisting of 79 protein-coding genes, 30 tRNA genes, and 4 rRNA genes. The difference is due to differences in counting duplicated genes in the inverted-repeat regions and annotation versions.
5. Choose at least eight protein-coding plastid genes from different functional groups. List each gene and briefly explain its biological function.
  - *psaA* – encodes a reaction-center protein of Photosystem I.
  - *psbA* – encodes a core protein of Photosystem II involved in photosynthesis.
  - *atpA* – encodes a subunit of ATP synthase involved in ATP production.
  - *petA* – encodes a cytochrome b6f complex protein involved in electron transport.
  - *rbcL* – encodes the large subunit of Rubisco involved in carbon fixation.
  - *rpoB* – encodes a subunit of the plastid RNA polymerase involved in transcription.
  - *rps12* – encodes a ribosomal protein involved in protein synthesis.
  - *ndhA* – encodes a subunit of the NADH dehydrogenase-like complex involved in electron transport.
6. Identify important RNA and RNA-processing features. Include the rRNA genes, examples of tRNA genes, and at least two genes with introns if present in your genome.
  - The chloroplast genome contains four major rRNA genes: ***rrn4.5***, ***rrn5***, ***rrn16***, and ***rrn23***, which are components of the ribosome and are important for protein synthesis. Examples of tRNA genes include *trnK-UUU*, *trnQ-UUG*, *trnS-GCU*, *trnR-UCU*, and *trnH-GUG*. Several genes contain introns, including *rps16*, *atpF*, *rpoC1*, *ycf3*, *clpP*, *petB*, *petD*, *rpl2*, *ndhB*, and *ndhA*.
7. Describe any pseudogenes, gene losses, duplications, rearrangements, or other unusual features reported for your plastid genome. If none are reported, state this clearly.
  - No pseudogenes are annotated in the current NCBI record. A notable feature is the duplication of genes within the inverted-repeat regions, including protein-coding, tRNA, and rRNA genes. The sources reviewed do not report a specific major gene loss or genome rearrangement in *Musa itinerans* chloroplast DNA.
8. What is the GC content of your plastid genome? Based on your Galaxy results and annotation, describe two other notable sequence or structural observations.
  - The overall GC content of the chloroplast genome is 36.82%. Two notable observations are that the genome has a typical quadripartite structure (LSC–IR–SSC–IR) and that the IR regions have a higher GC content (39.89%) than the LSC (35.13%) and SSC (30.71%) regions.

9. Compare plastid and mitochondrial genomes. Give at least five similarities and five differences, considering location, biological role, inheritance, genome organization, gene content, copy number, and evolutionary behavior.

- **Similarities:**

1. Both are organellar genomes.
2. Both originated through endosymbiosis.
3. Both contain their own genes.
4. Both occur in multiple copies within cells.
5. Both can be used in evolutionary and phylogenetic studies.

- **Differences:**

1. Plastid genomes are found in chloroplasts, while mitochondrial genomes are found in mitochondria.
2. Plastids contain genes associated with photosynthesis, while mitochondria contain genes mainly associated with cellular respiration.
3. Plastid genomes are generally more compact and structurally conserved.
4. Plant mitochondrial genomes are usually more structurally variable and can undergo rearrangements.
5. Plastid and mitochondrial genomes have different gene contents and genome organizations.

10. Explain the practical value of plastid genomes in research. List as many advantages as you can compared with the nuclear genome, including nuclear sex chromosomes where applicable, and also explain important limitations. Give one research question for which plastid data would be useful and one for which nuclear genomic data would be more appropriate.

- Plastid genomes are useful for species identification, DNA barcoding, phylogenetic analysis, population studies, biodiversity studies, and conservation research. They are advantageous because they are relatively small, compact, and conserved, making them easier to sequence and analyze than nuclear genomes. However, plastid genomes represent only the plastid lineage and therefore cannot provide the complete genetic history of an organism. They may also be affected by maternal or uniparental inheritance and hybridization.

### **Plastid vs Mitochondrial Genome Comparison**

Complete the table below using information from your selected organism and reliable references. If a feature varies among plant lineages, state that variation instead of forcing a single answer.

| <b>Feature</b>            | <b>Plastid genome</b>                                                   | <b>Mitochondrial genome</b>                                           |
|---------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Cellular location         | Chloroplasts and other plastids                                         | Mitochondria                                                          |
| Main biological functions | Photosynthesis, carbon fixation, and synthesis of some plastid proteins | Cellular respiration, energy production, and mitochondrial metabolism |

|                                   |                                                                                                                                                     |                                                                                                                            |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Typical genome organization       | Usually circular, with LSC–IR–SSC–IR quadripartite structure<br>Highly variable; may occur as circular, linear, branched, or multiple DNA molecules | Highly variable; may occur as circular, linear, branched, or multiple DNA molecules                                        |
| Relative genome size              | Usually smaller and more conserved; <i>Musa itinerans</i> has 168,985 bp                                                                            | Usually much larger and highly variable in plants                                                                          |
| Gene content                      | Mainly genes for photosynthesis, transcription, translation, rRNAs, and tRNAs                                                                       | Mainly genes for respiration, oxidative phosphorylation, translation, and some other mitochondrial functions               |
| Copy number                       | Usually present in multiple copies per cell/organelle                                                                                               | Multiple copies per cell/organelle; copy number varies among tissues and species                                           |
| Inheritance                       | Usually uniparental, commonly maternal in angiosperms                                                                                               | Usually uniparental, commonly maternal in angiosperms                                                                      |
| Recombination / structural change | Generally low recombination and relatively conserved structure, although some rearrangements occur                                                  | Frequent recombination and high structural variation; repeats can cause genome rearrangements                              |
| Mutation / substitution pattern   | Generally moderate and relatively conserved; often useful for phylogenetic studies                                                                  | Protein-coding sequences generally evolve slowly, but genome structure can change rapidly                                  |
| Common research applications      | DNA barcoding, species identification, phylogeny, plant evolution, biodiversity, and conservation                                                   | Studies of plant evolution, cytoplasmic inheritance, mitochondrial function, fertility, and mitochondrial genome evolution |